

Review Exercises with Solutions

Preparations

Loading packages

```
import pandas as pd
import matplotlib.pyplot as plt
import numpy as np
import statsmodels.api as sm
import statsmodels.formula.api as smf
import itertools
```

Loading dataset

The dataset is the hotels-europe dataset.

```
ROOT = r'C:\Users\PC_DS_ECON_5\Desktop\data-analytics-python'
hotels_europe = pd.read_parquet(ROOT + '/data/hotels_europe_restr_df.parquet')
```

Variables

```
hotels_europe.columns
```

```
Index(['hotel_id', 'city', 'distance', 'stars', 'rating', 'country',
      'accommodation_type', 'price', 'holiday', 'yearmonth'],
      dtype='str')
```

Variable	Description
hotel_id	Hotel identifier.
yearmonth	Month of observation.
city	City in which the hotel is located.
distance	Distance between the hotel and the city center (km).
stars	Hotel star rating.
rating	Average customer rating.
country	Country in which the hotel is located.
accommodation_type	Type of accommodation offered by the hotel.
price	Hotel price per night (USD).
holiday	Indicator for whether the observation refers to a holiday period.

Ex.1. Inspecting the variable `stars`

a. What does the `value_counts()` method do in each of the following cases?

Possible Answer

In the first case (`df1`), the `value_counts()` method counts the **number of occurrences** of each value of the variable `stars`, i.e. the **frequency** associated with each value of `stars`.

In the second case (`df2`), the `value_counts()` method computes the **share of observations** corresponding to each value of the variable `stars`, i.e. the **relative frequency** associated with each value of `stars`.

b. What does the `sort_values()` method do?

Possible Answer

It arranges the rows of the dataframe in ascending order according to the variable `stars`.

c. What does the `pd.merge()` method do in the current case?

Possible Answer

It merges `df1` and `df2` based on the variable `stars`.

d. In the dataframe `df3`, what does the column `count` correspond to?

Possible Answer

The column `count` contains the **frequency** associated with each value of the variable `stars`.

e. In the dataframe `df3`, what does the column `proportion` correspond to?

Possible Answer

The column `proportion` contains the **relative frequency** associated with each value of the variable `stars`.

```
df0= hotels_europe.copy()
df1= (
    df0['stars']
        .value_counts(normalize=False)
        .reset_index()
        .sort_values(by='stars', ascending=True)
        .reset_index(drop=True)
        .copy()
)
print('df1 columns:', df1.columns.to_list())
df2= (
    df0['stars']
        .value_counts(normalize=True)
        .reset_index()
        .sort_values(by='stars', ascending=True)
        .reset_index(drop=True)
        .copy()
```

```

)
print('df2 columns:', df2.columns.to_list())
df3= pd.merge(
    left=df1,
    right=df2,
    on='stars',
    how='outer'
)
df3

```

```

df1 columns: ['stars', 'count']
df2 columns: ['stars', 'proportion']

```

	stars	count	proportion
0	1.0	613	0.022651
1	1.5	45	0.001663
2	2.0	3391	0.1253
3	2.5	856	0.03163
4	3.0	9683	0.357795
5	3.5	2049	0.075712
6	4.0	8378	0.309574
7	4.5	355	0.013118
8	5.0	1693	0.062558

Ex. 2. Unit of observation

Based on the code and the corresponding output below, what is the unit of observation in the `hotels_europe` dataframe here? Briefly explain why.

- i. The unit of observation is `hotel_id`.
- ii. The unit of observation is `hotel_id × yearmonth`
- iii. The unit of observation is `hotel_id × yearmonth × holiday`

Possible Answer

- ii. The unit of observation is `hotel_id × yearmonth`

The combination of `hotel_id` and `yearmonth` uniquely identifies each observation, whereas `hotel_id` alone does not. There is exactly one observation for each `hotel_id × yearmonth` pair, but multiple observations for the same `hotel_id` in some cases.

Adding the dummy variable `holiday` is not necessary to uniquely identify the observations in this dataframe.

Side Note

The unit of observation is not just a mechanical or mathematical concept. It also needs to make sense in the context of the dataset. For example, other variables such as `price` might also uniquely identify observations. In practice, the unit of observation is usually defined by a combination of identifiers, time variables, categorical variables, or dummy variables that specify what entities are being observed and at what level the dataset provides information.

```
df0= hotels_europe.copy()
df1= (
    df0
        .value_counts(subset=['hotel_id'])
        .reset_index()
        .rename(columns={'count': 'n_obs'})
        .value_counts(subset=['n_obs'])
        .reset_index()
        .copy()
)
print('1:\n', df1, '\n', sep='')
df2= (
    df0
        .value_counts(subset=['hotel_id', 'yearmonth'])
        .reset_index()
        .rename(columns={'count': 'n_obs'})
        .value_counts(subset=['n_obs'])
        .reset_index()
        .copy()
)
print('2:\n', df2, '\n', sep='')
df3= (
    df0
        .value_counts(subset=['hotel_id', 'yearmonth', 'holiday'])
        .reset_index()
```

```
.rename(columns={'count': 'n_obs'})
.value_counts(subset=['n_obs'])
.reset_index()
.copy()
)
print('3:\n', df3, '\n', sep='')
```

```
1:
  n_obs  count
0      2  13898
1      1   5446
```

```
2:
  n_obs  count
0      1  33242
```

```
3:
  n_obs  count
0      1  33242
```

Ex. 3. Transforming the variable stars

- a. Explain the purpose of the code below.

Possible Answer

The purpose of the code is to transform the numerical variable `stars` into an ordered categorical variable.

- b. What is the new variable `stars_cat`?

Possible Answer

It contains the exact same information as `stars`, but it is explicitly stored as an ordered categorical variable rather than as a numerical variable.

- c. Why are `drop_duplicates()` and `dropna()` used?

Possible Answer

Together, the two methods extract the unique non-missing values of the variable `stars`. `drop_duplicates()` keeps only one occurrence of each distinct value, and `dropna()` removes the missing-value placeholder from the resulting list.

Side Note

It is legitimate to ask: What is the purpose of all this? Treating hotel stars as a numerical variable may be inappropriate in some contexts because the numerical distances between categories do not necessarily reflect equal differences in the underlying characteristics. An ordered categorical variable such as `stars_cat` is also useful when fitting a regression model with dummy variables for each hotel star category.

```
df0= hotels_europe.copy()
print('stars variable datatype:', df0['stars'].dtype)

stars_unique_values= (
    df0['stars']
        .drop_duplicates()
        .dropna()
        .sort_values(ascending=True)
        .to_list()
        .copy()
)
print('stars_unique_values:', stars_unique_values)
df0['stars_cat'] = pd.Categorical(
    values=df0['stars'],
    categories=stars_unique_values,
    ordered=True
)

hotels_europe= df0.copy()
```

```
stars variable datatype: Float64
stars_unique_values: [1.0, 1.5, 2.0, 2.5, 3.0, 3.5, 4.0, 4.5, 5.0]
```

Ex. 4. Inspecting the variable distance

- a. In the code below, what is the purpose of the line with `.loc`?

i Possible Answer

The purpose of the line with `.loc` is to select the subset of observations with `yearmonth == '2017-11-01'`. Put differently, the purpose is to keep all observations from November 2017.

- b. What kind of plot does the code below produce?

i Possible Answer

A histogram.

- c. How would you label the vertical (y-)axis?

i Possible Answer

Relative Frequency, Share of Observations, or Proportion of Observations

- d. Is the distribution of `distance` symmetric? Would you expect the median of `distance` or the mean of `distance` to be higher?

i Possible Answer

The distribution is clearly not symmetric but right-skewed. Therefore, the mean of `distance` is likely higher than the median.

```
binwidth = 1
df0= hotels_europe.copy()

df0 = df0.loc[df0['yearmonth'].eq('2017-11-01'),:].copy()

max_distance = df0['distance'].max()
max_upper_bound = np.ceil(max_distance)

df0['distance_interval']= pd.cut(
    df0['distance'],
    bins=np.arange(
        start=0,
        stop=max_upper_bound + binwidth,
        step=binwidth),
    include_lowest=True
)

df0['distance_interval_mid'] = (
    df0['distance_interval']
    .map(lambda x: x.mid)
)

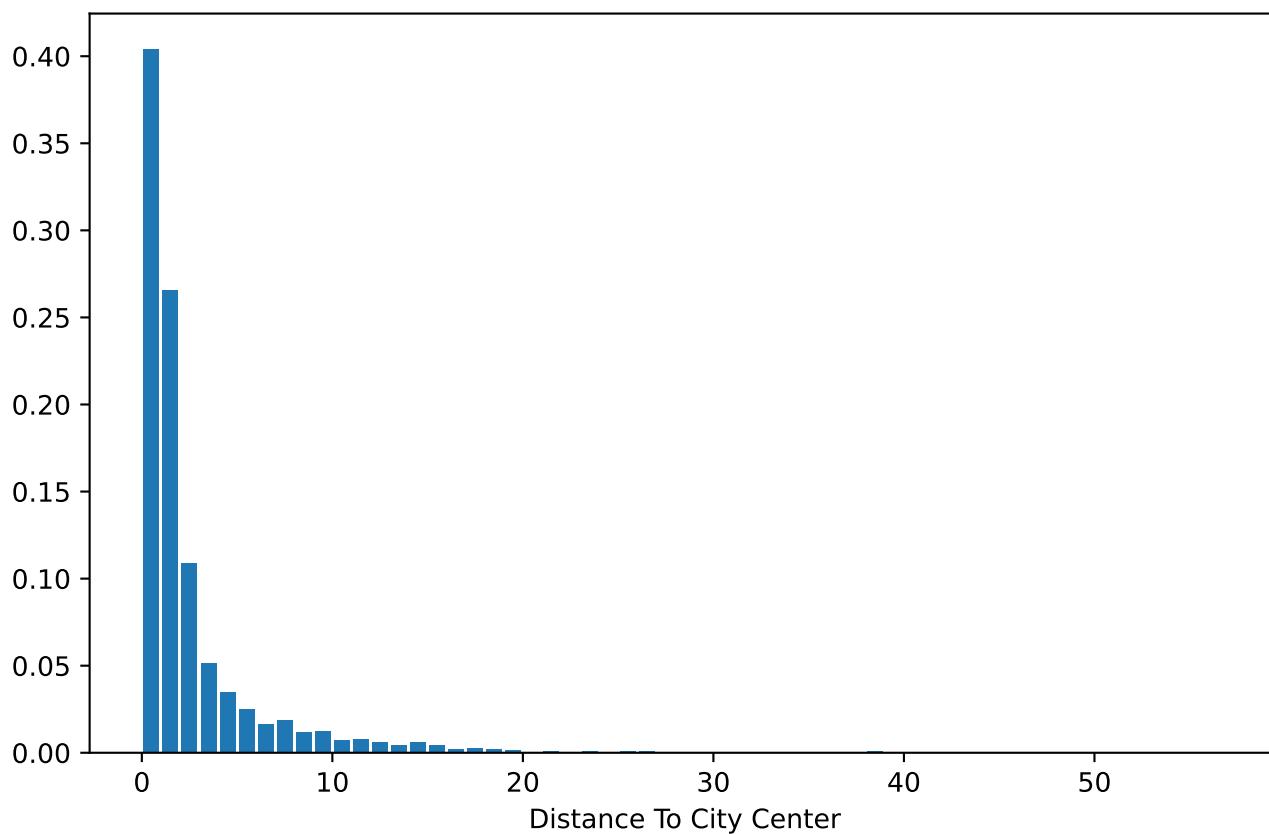
df1= (
```

```

df0
    .value_counts(subset=['distance_interval_mid'], normalize=True)
    .reset_index()
    .sort_values(by=['distance_interval_mid'], ascending=True)
    .reset_index(drop=True)
)

fig, ax = plt.subplots()
ax.bar(
    x=df1['distance_interval_mid'],
    height=df1['proportion'])
ax.set_xlabel('Distance To City Center')
plt.show()

```



Ex. 5. Defining distance categories

Briefly explain the purpose of the following code.

Possible Answer

The purpose of the code is to create an ordered categorical variable `distance_cat` based on the continuous variable `distance`, where the categories correspond to specified distance intervals.

The first step transforms the numerical variable into an ordered categorical variable with distance intervals as categories.

The second step replaces the interval categories with simpler string labels that are more suitable for plotting.

```
binwidth = 1
df0= hotels_europe.copy()
max_upper_bound = np.ceil(df0['distance'].max() / 10) * 10
interval_bounds = [0, 1, 2, 3, 4, 6, 8, 10, 20, max_upper_bound]

df0['distance_cat_int']= pd.cut(
    df0['distance'],
    bins=interval_bounds,
    include_lowest=True
)

print('distance_cat_int:\n', repr(df0['distance_cat_int'].dtype))

distance_cat_list = (
    df0['distance_cat_int']
    .dropna()
    .drop_duplicates()
    .sort_values()
    .to_list()
)

distance_cat_dict = {
    iv: (
        f'({int(iv.left)}, {int(iv.right)})'
        if int(iv.left) > 0
        else f'[{int(iv.left)}, {int(iv.right)}]'
    )
    for iv in distance_cat_list
}

df0['distance_cat'] = (
    df0['distance_cat_int']
    .map(lambda x: distance_cat_dict[x])
)

distance_cat_unique = [cat for iv, cat in distance_cat_dict.items()]

df0['distance_cat'] = pd.Categorical(
    values=df0['distance_cat'],
    categories=distance_cat_unique,
```

```

        ordered=True
    )

print(
    'distance_cat:\n',
    repr(df0['distance_cat'].dtype)
)

hotels_europe = df0.copy()

```

```

distance_cat_int:
    CategoricalDtype(categories=[(-0.001, 1.0],    (1.0, 2.0],    (2.0, 3.0],    (3.0, 4.0],
                                (4.0, 6.0],    (6.0, 8.0],    (8.0, 10.0], (10.0, 20.0],
                                (20.0, 60.0]],
    , ordered=True, categories_dtype=interval[float64, right])
distance_cat:
    CategoricalDtype(categories=['[0, 1]', '(1, 2]', '(2, 3]', '(3, 4]', '(4, 6]', '(6, 8]',
                                '(8, 10]', '(10, 20]', '(20, 60]'],
    , ordered=True, categories_dtype=str)

```

Ex. 6. Descriptive statistics

- a. In the code below, what is the purpose of the line with `.loc`?

Possible Answer

The purpose of `.loc` is to keep only the observations for which `price` is not missing.

- b. For each of the statistics `statistic[1]`, `statistic[2]`, `statistic[3]`, `statistic[4]`:
- Explain what the statistic corresponds to.
 - Whether it is a measure of central tendency or a measure of dispersion.

Possible Answer

`statistic[1]` is the mean: the sum of all observed values of `price` divided by the number of observations. The mean is a measure of central tendency.

`statistic[2]` is the variance: the mean of squared deviations from the mean. The variance is a measure of dispersion.

`statistic[3]` is the standard deviation: the square root of the variance. The standard deviation is a measure of dispersion.

`statistic[4]` is the median: the 0.5-quantile. About 50% of the observations have a price less than or equal to the median. The median is a measure of central tendency.

```
df0 = hotels_europe.copy()
df0 = df0.loc[df0['price'].notna()], :].copy()

statistic = {}

statistic[1] = (
    df0['price'].sum()/df0['price'].shape[0]
)

statistic[2] = (
    ((df0['price'] - df0['price'].mean()) ** 2)
    .mean()
)

statistic[3] = (
    np.sqrt(
        ((df0['price'] - df0['price'].mean()) ** 2)
        .mean()
    )
)

statistic[4] = (
    df0['price'].quantile(0.5)
)

print(
    f'statistic 1: {statistic[1]:.1f}\n',
    f'statistic 2: {statistic[2]:.1f}\n',
    f'statistic 3: {statistic[3]:.1f}\n',
```

```
f'statistic 4: {statistic[4]:.1f}',  
sep=' '  
)
```

```
statistic 1: 118.2  
statistic 2: 11434.2  
statistic 3: 106.9  
statistic 4: 93.0
```

Ex. 7. Relationship between price and distance in three major cities

- a. Explain the purpose of the use of the two `.loc` in the code below.

i Possible Answer

The purpose of the first `.loc` is to keep only the observations that satisfy all the following criteria: - with `yearmonth == '2017-11-01'`, i.e. from November 2017 - `city` is either Paris, London, or Rome - `distance` is less than or equal to 10 km - `distance` is not missing - `price` is not missing
The second use of `.loc` selects only the observations corresponding to one city at a time so that each city can be plotted separately.

- b. Based on the code and the output below, holding city constant, does price appear to be mean independent from distance? Formally, does it appear to be the case that $\mathbb{E}[\text{price} \mid \text{distance}, \text{city}] \approx \mathbb{E}[\text{price} \mid \text{city}]$? Briefly justify your answer.

i Possible Answer

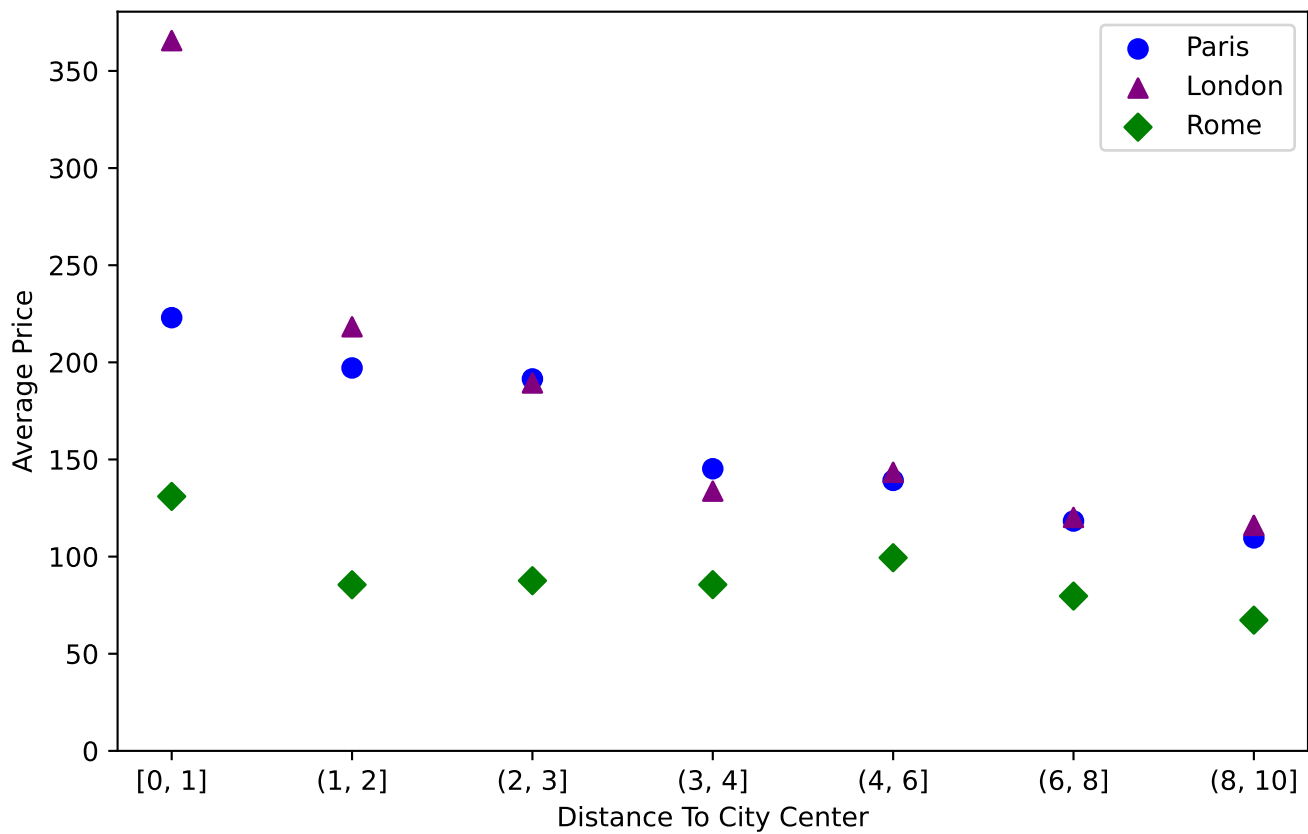
No. For each of the three cities, there appears to be a reasonably clear negative relationship between the mean price and the distance to the city centre.

```
df0 = hotels_europe.copy()
cities = ['Paris', 'London', 'Rome']
colours = ['blue', 'purple', 'green']
markers = ['o', '^', 'D']
df0 = df0.loc[df0['yearmonth'].eq('2017-11-01') &
              df0['city'].isin(cities) &
              df0['distance'].le(10) &
              df0['distance'].notna() &
              df0['price'].notna(),:].copy()

df1 = (
    df0
    .groupby(by=['distance_cat', 'city'])
    .agg(mean_price_by_dist_and_city = ('price', 'mean'))
    .reset_index()
    .sort_values(by=['city', 'distance_cat'])
    .reset_index(drop=True)
    .copy()
)

fig, ax = plt.subplots()
for i, city in enumerate(cities):
    df2 = df1.loc[df1['city'].eq(city),:].copy()
    ax.scatter(
        x=df2['distance_cat'].astype(str),
        y=df2['mean_price_by_dist_and_city'],
        label=city,
        marker=markers[i],
        color=colours[i],
        s=50
    )
```

```
ax.set_ylabel('Average Price')
ax.set_xlabel('Distance To City Center')
ax.set_ylim(bottom=0)
ax.legend()
```



Saving selected data

Saving the selected dataset as `hotels_selected`.

```
hotels_selected = df0.copy()
```

Ex. 8. Linear regression of price on distance for three major cities

Consider the following linear regression implemented in the code below:

$$\begin{aligned} \text{price} = & \beta_1 \cdot \text{distance} + \beta_{0, \text{London}} \cdot \mathbb{1}[\text{city} = \text{London}] + \\ & + \beta_{0, \text{Paris}} \cdot \mathbb{1}[\text{city} = \text{Paris}] + \beta_{0, \text{Rome}} \cdot \mathbb{1}[\text{city} = \text{Rome}] \end{aligned}$$

What is the interpretation of each of the coefficients $\beta_1, \beta_{0, \text{London}}, \beta_{0, \text{Paris}}, \beta_{0, \text{Rome}}$?

i Possible Answer

$\hat{\beta}_1 \approx -17.35$ means that holding city constant, hotels one kilometer further from the city centre tend to have prices that are about 17 USD lower on average.

$\hat{\beta}_{0, \text{London}} \approx 250$ means that the predicted average price per night in the city centre ($\text{distance}=0$) of London is 250 USD.

$\hat{\beta}_{0, \text{Paris}} \approx 230$ means that the predicted average price per night in the city centre ($\text{distance}=0$) of Paris is 230 USD.

$\hat{\beta}_{0, \text{Rome}} \approx 130$ means that the predicted average price per night in the city centre ($\text{distance}=0$) of Rome is 130 USD.

```
df0 = hotels_selected.copy()
cities = ['Paris', 'London', 'Rome']
colours = ['blue', 'purple', 'green']
markers = ['o', '^', 'D']
styles = ['solid', 'dotted', 'dashed']
df0 = (
    df0[['price', 'distance', 'city', 'rating']]
    .dropna()
    .reset_index(drop=True)
    .copy()
)

reg = smf.ols(formula='price ~ 0 + distance + city', data=df0).fit()
print(reg.params)
beta_1 = reg.params['distance']

fig, ax = plt.subplots()
for i, city in enumerate(cities):
    df2 = df0.loc[df0['city'].eq(city),:].copy()
    intercept = reg.params['city[' + city + ']']
    ax.scatter(
        x=df2['distance'],
        y=df2['price'],
        label=city,
        marker=markers[i],
        color=colours[i],
        s=10,
        alpha=0.05
    )
distance_values = np.linspace(start =0, stop=10, num= 1000)
```

```

price_hat = intercept + beta_1 * distance_values
ax.plot(
    distance_values,
    price_hat,
    color='black',
    label=f'price_hat = {reg.params['distance']:.2f} * distance + {intercept:.0f}',
    linestyle=styles[i],
    linewidth= 2
)

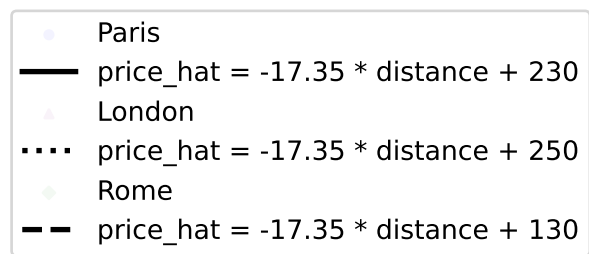
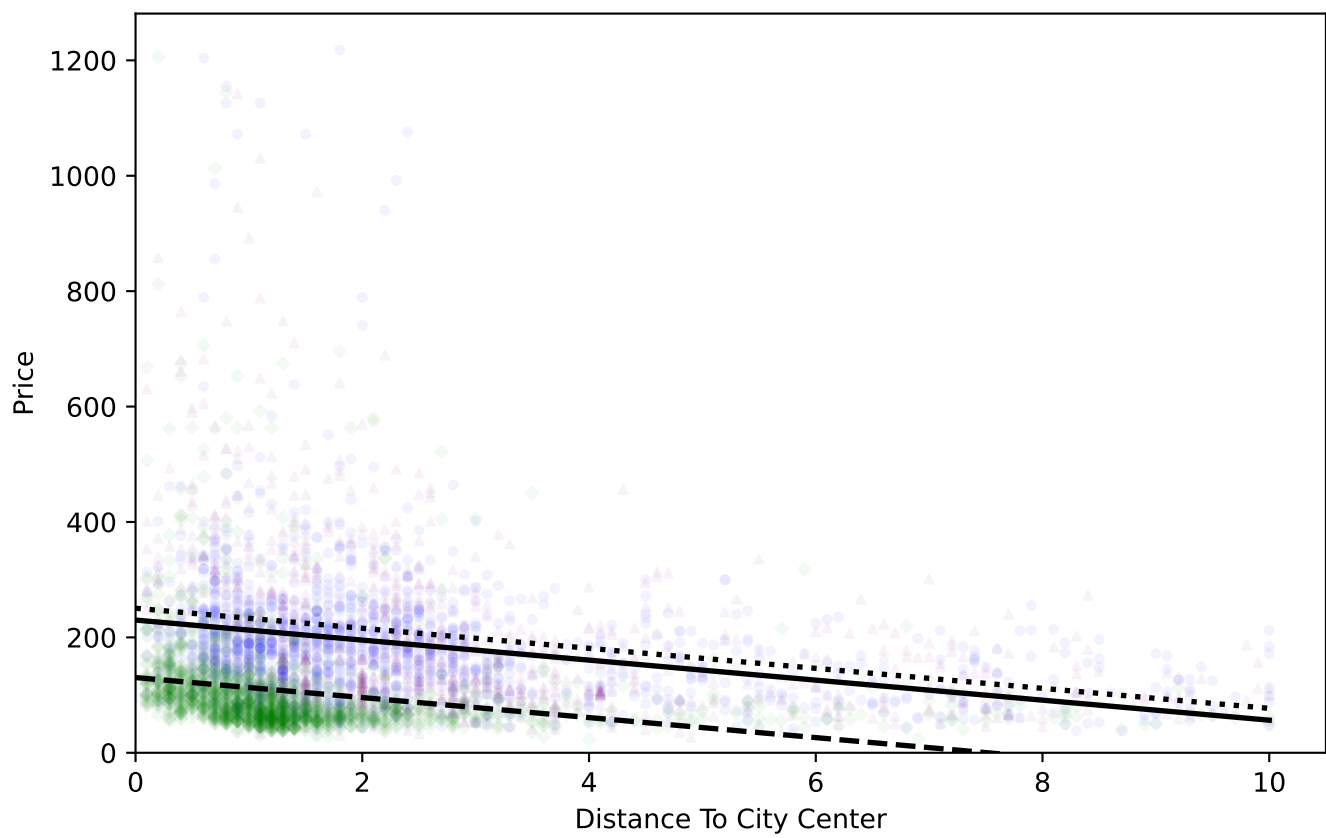
ax.set_ylabel('Price')
ax.set_xlabel('Distance To City Center')
ax.set_ylim(bottom=0)
ax.set_xlim(left=0)
ax.legend(
    bbox_to_anchor=(0.5,-0.5),
    loc='center'
)

```

```

city[London]    250.469641
city[Paris]     229.993839
city[Rome]      130.401668
distance        -17.351616
dtype: float64

```

Centering / demeaning the variable rating

The variable `rating_centered` is defined as `rating` minus the mean of `rating`:

$$rating_centered_i = rating_i - \overline{rating}$$

```
df0 = hotels_selected.copy()
df0['rating_centered'] = df0['rating'] - df0['rating'].mean()
hotels_selected = df0.copy()
```

Ex. 9. Adding (centered) hotel ratings to the linear regression of price on distance for three major cities

Consider the following three regressions implemented by the code below.

Explain what the two identical values printed at the end of the code represent.

Possible Answer

The two values represent the omitted variable bias in the estimated coefficient of **distance** (the regressor of interest) that arises from omitting **rating_centered** (demeaned hotel ratings) from the regression. In both cases, the other control variables (city dummies) remain the same.

As expected, the difference between the two coefficient estimates is equal to the product of:

- the estimated coefficient of **rating_centered** (the control variable) in the regression that includes it, and
- the estimated coefficient of **distance** in the auxiliary regression of **rating_centered** (the omitted control variable) on **distance** and the other control variables (city dummies).

```
df0 = hotels_selected.copy()
cities = ['Paris', 'London', 'Rome']

df0 = (
    df0[['price', 'distance', 'city', 'rating_centered']]
        .dropna()
        .reset_index(drop=True)
        .copy()
)

include_rating = smf.ols(
    formula='price ~ distance + rating_centered + city',
    data=df0
).fit()
omit_rating = smf.ols(
    formula='price ~ distance + city',
    data=df0
).fit()
auxiliary = smf.ols(
    formula='rating_centered ~ distance + city',
    data=df0
).fit()

print(
    'Include rating:\n',
    include_rating.params[['distance', 'rating_centered']],
    '\n\n',
    'Omit rating:\n',
    omit_rating.params[['distance']],
    '\n\n',
    'Auxiliary:\n',
    auxiliary.params[['distance']],
    '\n\n',
    'Difference:\n',
```

```
omit_rating.params['distance'] - include_rating.params['distance'],  
'\n\n',  
'Product:\n',  
include_rating.params['rating_centered'] * auxiliary.params['distance'],  
sep='')  
)
```

Include rating:

```
distance      -13.565516  
rating_centered  67.620771  
dtype: float64
```

Omit rating:

```
distance      -17.351616  
dtype: float64
```

Auxiliary:

```
distance      -0.05599  
dtype: float64
```

Difference:

```
-3.786099518130273
```

Product:

```
-3.7860995181300505
```

Ex. 10. Including distance as categorical variable in the linear regression of price on distance for three major cities

Consider the following linear regression:

$$\begin{aligned} \text{price} = & \beta_0 + \sum_{j \in \text{DistanceCategories}} \beta_j \cdot \mathbb{1}[\text{DistanceCategory} = j] + \\ & + \gamma \cdot \text{rating_centered} + \delta_{\text{Paris}} \cdot \mathbb{1}[\text{city} = \text{Paris}] + \delta_{\text{Rome}} \cdot \mathbb{1}[\text{city} = \text{Rome}] \end{aligned}$$

- a. What is the interpretation of the intercept coefficient $\hat{\beta}_0$ / **Intercept**?

i Possible Answer

$\hat{\beta}_0 \approx 256$ means that the predicted average price for a hotel in London (the reference city) with an average rating (**rating_centered** = 0) and within a 1 km radius of the city centre (the reference distance category) is approximately 256 USD.

- b. What is the interpretation of coefficient $\hat{\beta}_{(3,4]}$ / **distance_cat[T.(3, 4]]**?

i Possible Answer

$\hat{\beta}_{(3,4]} \approx -72.3$ means that, holding city and hotel rating constant, the predicted average price for hotels located between 3 and 4 kilometres from the city centre is approximately 72 USD lower than for hotels within a 1 km radius of the city centre (the reference distance category).

- c. What is the interpretation of coefficient $\hat{\delta}_{\text{Rome}}$ / **city[T.Rome]**?

i Possible Answer

$\hat{\delta}_{\text{Rome}} \approx -138$ means that, holding distance category and hotel rating constant, the predicted average price for hotels in Rome is approximately 138 USD lower than for hotels in London (the reference city).

```
df0 = hotels_selected.copy()
cities = ['Paris', 'London', 'Rome']
markers = ['o', '^', 'D']

df0['distance_cat'] = df0['distance_cat'].cat.remove_unused_categories()
df0 = (
    df0[['price', 'distance_cat', 'city', 'rating_centered']]
        .dropna()
        .reset_index(drop=True)
        .copy()
)

reg = smf.ols(
    formula='price ~ distance_cat + rating_centered + city',
    data=df0
).fit()

print(reg.params)
```

```

df1 = df0.copy()
df1['rating_centered'] = 0
df1['price_hat'] = reg.predict(df1)

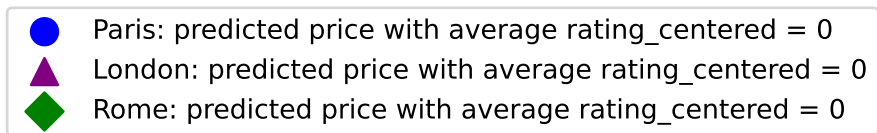
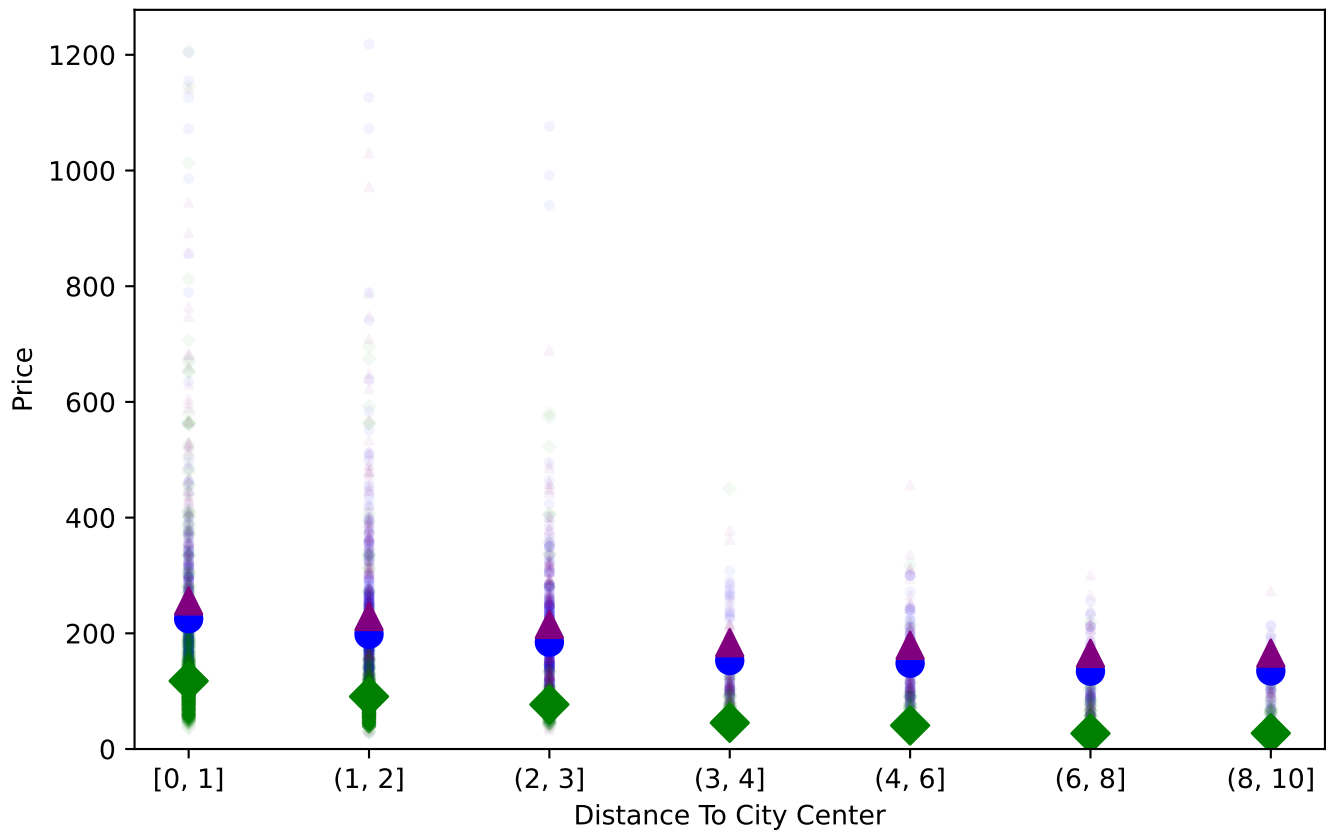
fig, ax = plt.subplots()
for i, city in enumerate(cities):
    df2 = df1.loc[df1['city'].eq(city),:].copy()
    df2 = (
        df2
        .sort_values(by=['distance_cat'])
        .reset_index(drop=True)
        .copy()
    )
    ax.scatter(
        x=df2['distance_cat'],
        y=df2['price'],
        marker=markers[i],
        color=colours[i],
        s=10,
        alpha=0.05
    )
for i, city in enumerate(cities):
    df2 = df1.loc[df1['city'].eq(city),:].copy()
    df2 = (
        df2[['distance_cat', 'price_hat', 'city']]
        .drop_duplicates()
        .sort_values(by=['distance_cat'])
        .reset_index(drop=True)
        .copy()
    )
    ax.scatter(
        df2['distance_cat'],
        df2['price_hat'],
        label=city + ': predicted price with average rating_centered = 0',
        color=colours[i],
        s=100,
        marker=markers[i]
    )

ax.set_ylabel('Price')
ax.set_xlabel('Distance To City Center')
ax.set_ylim(bottom=0)
ax.legend(
    bbox_to_anchor=(0.5, -0.5),
    loc='center'
)

```

Intercept	256.088672
distance_cat[T.(1, 2)]	-26.941947
distance_cat[T.(2, 3)]	-40.666527
distance_cat[T.(3, 4)]	-72.282671

```
distance_cat[T.(4, 6]]      -76.892858
distance_cat[T.(6, 8]]      -90.705536
distance_cat[T.(8, 10]]     -90.291501
city[T.Paris]                -30.850610
city[T.Rome]                 -138.452249
rating_centered              65.470361
dtype: float64
```



Ex. 11. Confidence interval for the price-distance relationship

The code below: - simulates the bootstrap distribution of the coefficient estimate $\hat{\beta}_{(3,4]}$ / `distance_cat[T.(3, 4)]`, - plots the simulated bootstrap distribution, - and calculates selected quantiles of this distribution.

- a. Based on the bootstrap percentile method, how would you construct a symmetric 95% confidence interval for the regression coefficient? Which quantile would be the lower bound and which quantile would be the upper bound?

i Possible Answer

For a symmetric 95% confidence interval based on the bootstrap percentile method: - the lower bound is the 2.5th percentile of the bootstrap distribution: -82.6 - the upper bound is the 97.5th percentile of the bootstrap distribution: -62.0

Then, about 2.5% of the simulated coefficient values fall below the lower bound and about 2.5% fall above the upper bound. Therefore, approximately 95% of the simulated bootstrap estimates lie between these two bounds.

- b. What is the reason behind constructing a confidence interval instead of simply interpreting the estimated coefficient as the true regression coefficient?

i Possible Answer

We only observe one sample of data. If we repeatedly drew different random samples from the same population, the estimated coefficient would vary. The confidence interval reflects the sampling uncertainty in the estimated coefficient.

i Side Note

The population could actually be a real / existing population (all hotels in the three cities within 10km from the city centre) from which our data were drawn. But the population could also be thought of as the unknown statistical process that generated the data, i.e. the process through which hotel prices are determined. It could happen that we observe all hotels in the three cities within 10km from the city centre and we still call our data a sample, in which case our population is theoretical in nature. The statistical population can be real (existing) or theoretical. In the latter case, we mean by population the underlying unknown complex joint distribution of random variables.

- c. Which familiar probability distribution does the shape of the bootstrap distribution resemble?

i Possible Answer

The bootstrap distribution looks like a normal distribution. It appears symmetric and bell-shaped.

- d. What would be another approach to constructing a confidence interval for the regression coefficient? Which theorem can be used to justify this approach? What does the theorem say?

i Possible Answer

Another approach would be analytical (i.e. without simulation) using large sample theory. The theorem that could be used is the Central Limit Theorem (CLT). In its simplest form, the CLT says that the distribution of the sample mean looks more and more like a normal distribution as the sample size increases. In sufficiently large samples, the distribution of the sample mean can be well approximated by a normal

distribution. The CLT also applies in the context of OLS linear regression coefficients. Under suitable assumptions, the distribution of OLS coefficient estimators is well approximated by a normal distribution in large samples.

```
df0= hotels_selected.copy()

df0= (
    df0[['price', 'distance_cat', 'city', 'rating_centered']]
    .dropna()
    .reset_index(drop=True)
    .copy()
)

reg= smf.ols(
    formula='price ~ distance_cat + rating_centered + city',
    data=df0
).fit()

point_estimate= reg.params['distance_cat[T.(3, 4)']']

rng= np.random.default_rng(946372)

R= 10000
N= df0.shape[0]
beta_sim= np.empty(R)
for i in range(R):
    bootstrap_indices= rng.choice(
        a=N,
        size=N,
        replace=True
    )

    dfr= df0.iloc[bootstrap_indices]

    reg_bootstrap= smf.ols(
        formula='price ~ distance_cat + rating_centered + city',
        data=dfr
    ).fit()
    beta_sim[i]= reg_bootstrap.params['distance_cat[T.(3, 4)']']

dfs = pd.DataFrame({'beta_sim': beta_sim})
beta_sim_min = dfs['beta_sim'].min()
beta_sim_max = dfs['beta_sim'].max()
lower_bound_min = np.floor(beta_sim_min)
upper_bound_max = np.ceil(beta_sim_max)
binwidth = 2
bins = np.arange(
    start=lower_bound_min,
    stop=upper_bound_max + binwidth,
    step=binwidth
)
```

```

dfs['beta_sim_int'] = pd.cut(
    x=dfs['beta_sim'],
    bins=bins,
    include_lowest=True
)

dfs['beta_sim_mid'] = (
    dfs['beta_sim_int']
    .map(lambda x: x.mid)
)

dfd = (
    dfs['beta_sim_mid']
    .value_counts(normalize=True)
    .reset_index()
    .sort_values(by=['beta_sim_mid'], ascending=True)
    .reset_index(drop=True)
    .copy()
)

fig, ax = plt.subplots()
ax.bar(
    x=dfd['beta_sim_mid'],
    height=dfd['proportion'],
    width=0.8 * binwidth,
    label='Simulated Bootstrap Distribution'
)
ax.axvline(
    x=point_estimate,
    c='red',
    linewidth=5,
    label=f'Point estimate: {point_estimate:.1f}'
)
ax.set_ylabel('Relative Frequency')
ax.set_xlabel('Coefficient Value')
ax.legend(
    bbox_to_anchor=(0.5, -0.5),
    loc='center'
)

print(
    '\n1% Quantile:',
    f"{dfs['beta_sim'].quantile(0.01):.1f}",
    '\n2.5% Quantile:',
    f"{dfs['beta_sim'].quantile(0.025):.1f}",
    '\n5% Quantile:',
    f"{dfs['beta_sim'].quantile(0.05):.1f}",
    '\n95% Quantile:',
    f"{dfs['beta_sim'].quantile(0.95):.1f}",
    '\n97.5% Quantile:',
    f"{dfs['beta_sim'].quantile(0.975):.1f}",

```

```

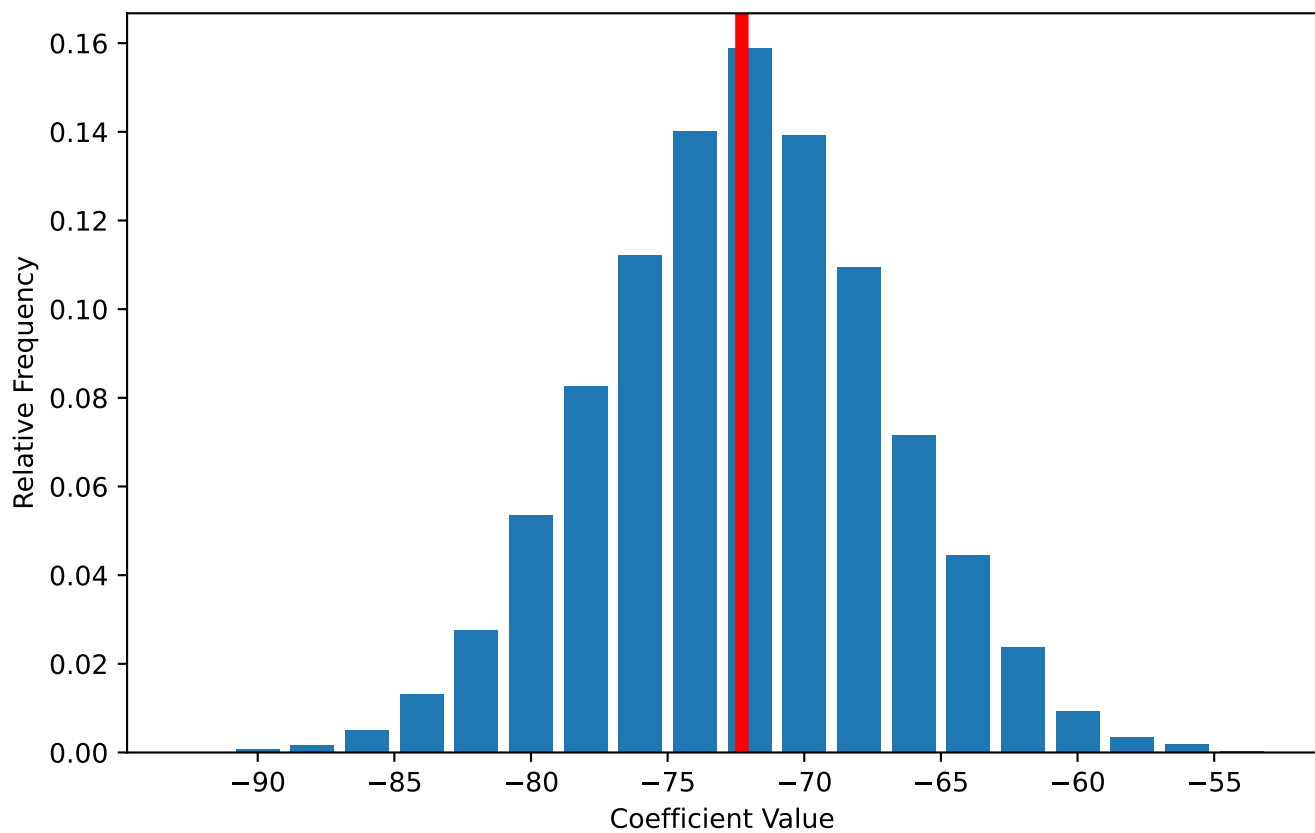
'\n99% Quantile:',
f"{dfs['beta_sim'].quantile(0.99):.1f}",
sep=' '
)

```

```

1% Quantile:-84.4
2.5% Quantile:-82.6
5% Quantile:-80.9
95% Quantile:-63.6
97.5% Quantile:-62.0
99% Quantile:-60.1

```



█ Point estimate: -72.3
█ Simulated Bootstrap Distribution